STPM Physics · Semester 2 · Chapter 15

Direct Current Circuits

15.1 Internal Resistance

Definitions & laws

- **Electromotive force (e.m.f.)** of a source is the energy converted from other forms into electrical energy per unit charge passing through the source; it equals the terminal p.d. when no current flows.
- **Internal resistance** of a source is the resistance to the flow of current within the source itself, arising from the materials of the cell.
- **Terminal potential difference** is the p.d. across the terminals of a source when it delivers current; it is less than the e.m.f. by the 'lost volts' Ir.
- **Maximum power theorem:** the power delivered to the external resistor is maximum when the external resistance equals the internal resistance of the source, R = r.
- **Cells in series** have a total e.m.f. and total internal resistance equal to the sums of the individual values (for cells in the same sense).

Formulae

Quantity / law	Formula	Symbols, units & conditions
E.m.f. equation	$E = V + Ir = I(R + r)$	E = e.m.f. (V); V = terminal p.d. (V); I = current (A); r = internal resistance (Ω); R = external resistance (Ω)
Terminal p.d.	$V = E - Ir$	A graph of V against I is a straight line with intercept E and gradient −r
Internal resistance	$r = \frac{\Delta V}{\Delta I}$; $r = \frac{E - V}{I}$	$\frac{\Delta V}{\Delta I}$ from two readings of the terminal p.d. (magnitude of the gradient of the V–I graph)
Power in external resistor	$P = I^2R = \frac{E^2R}{(R + r)^2}$	P = power in the external resistor (W)
Maximum power transfer	$R = r$; $P_{max} = \frac{E^2}{4r}$	Power in the external resistor is a maximum when R = r
Power dissipated in the source	$P_{int} = I^2r$	P_{int} = power dissipated in the internal resistance (W)
Total power supplied	$P = EI$	P = total power supplied by the source (W)

How to answer (MPM style)

- State E = V + Ir and substitute; find I first from $I = \frac{E}{R + r}$ when the circuit values are given.

- For two readings, write two simultaneous equations $E = V_1 + I_1r$ and $E = V_2 + I_2r$, subtract to find r, then find E.
- For a V-I graph state: intercept on the V-axis = E, gradient = −r.
- Describe terminal p.d. as less than e.m.f. because of the p.d. across the internal resistance when a current flows.

Section A traps

Trap: Taking the terminal p.d. as equal to the e.m.f. when the cell is delivering current; V = E only on open circuit.

Trap: Defining internal resistance as $\frac{E}{I}$ or $\frac{V}{I}$; these are the total and external resistances, respectively.

Trap: Adding only the external resistance when finding I, ignoring r; the correct total is R + r.

Trap: Forgetting that a series battery of N cells also has N times the internal resistance.

Tricks

Trick: A voltmeter reading before the switch is closed gives E; the reading afterwards gives V. Then $r = \frac{E - V}{I}$.

Trick: For maximum power, the external and internal resistances are equal, so half of the total power is dissipated in the source (efficiency 50%).

15.2 Kirchhoff's Laws

Definitions & laws

- **Kirchhoff's first law:** the sum of the currents entering a junction equals the sum of the currents leaving the junction. It is based on the conservation of charge.
- **Kirchhoff's second law:** in any closed loop the algebraic sum of the e.m.f.s equals the algebraic sum of the products IR (potential drops) in the loop. It is based on the conservation of energy.
- **Sign convention:** choose a direction for each current and for the loop; an e.m.f. is positive if the loop direction passes through the cell from the negative to the positive terminal, and IR is positive if the loop direction is the same as the current direction.
- **Opposing cells** in a loop give a net e.m.f. equal to the difference of the e.m.f.s.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Kirchhoff's first law (junction rule)	$\Sigma I_{in} = \Sigma I_{out}$	I = current (A); applies at a junction
Kirchhoff's second law (loop rule)	$\Sigma E = \Sigma IR$	E = e.m.f. (V); I = current (A); R = resistance (Ω); applies around a closed loop
Resistors in series	$R = R_1 + R_2 + \ldots$	R = total resistance (Ω)
Resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$	R = total resistance (Ω)
Resistor with a voltmeter across it	$\frac{RR_V}{R + R_V}$	R_V = finite resistance of the voltmeter (Ω); R = resistor (Ω); combined resistance when the voltmeter is connected across R

How to answer (MPM style)

- Draw the circuit with labelled currents and a chosen loop direction before writing the equations.
- Write the junction equation and one loop equation per independent loop, then solve the simultaneous equations.
- State the direction of each current found; a negative answer means the current flows opposite to the direction assumed.
- For a real voltmeter, replace the resistor and voltmeter by their parallel equivalent before applying the potential divider rule.

Section A traps

Trap: Adding the e.m.f.s of opposing cells; the net e.m.f. is the difference, for example 12 V − 4.0 V = 8.0 V.

Trap: Ignoring the voltmeter resistance in a high-resistance circuit; a voltmeter of finite resistance reduces the p.d. it measures.

Trap: Writing the junction law with all currents on the same side but with mixed signs; check that the total entering equals the total leaving.

Trap: Quoting conservation of charge for the loop law; the loop law comes from conservation of energy.

Tricks

Trick: At a junction, total in minus known out gives the unknown current, with direction determined by the sign of the result.

> **Trick:** A loop with only two cells and resistors in series: $I = \frac{\text{net e.m.f.}}{\text{total resistance}}$; first decide which cells oppose.

15.3 Potential Divider

Definitions & laws

- **A potential divider** is a circuit in which two or more resistors in series are connected across a supply so that the supply voltage is shared in proportion to the resistances; the output is taken across one of them.
- **Output voltage** is directly proportional to the resistance across which it is taken, so the larger resistor receives the larger share.
- **Loading effect:** a load connected in parallel with the output resistor reduces the effective resistance of that arm, so the output p.d. decreases below the no-load value.
- **Sensors:** the resistance of a thermistor (NTC) decreases as temperature increases, and the resistance of an LDR decreases as light intensity increases; the p.d. across the sensor in a divider therefore decreases.
- **A rheostat as a potential divider** has its whole track across the supply, which allows the output to vary continuously from zero to the supply p.d.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Potential divider output	$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$	V_{out} = output p.d. (V); V_{in} = input p.d. (V); R_2 = resistor across which the output is taken (Ω); R_1 = other resistor (Ω)
Ratio of output to input	$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2}$	Ratio is dimensionless
Voltage ratio in series	$V_1 : V_2 = R_1 : R_2$	For resistors in series
Divider with a load	$R_2 \rightarrow \frac{R_2 R_L}{R_2 + R_L}$	R_L = load resistance (Ω); replace R_2 by this combination in the formula
Current in the divider	$I = \frac{V_{in}}{R_1 + R_2}$; $V = IR$	I = current (A); V = p.d. across each resistor (V), using its own R

How to answer (MPM style)

- Identify the resistor across which the output is taken, then write Vout = Vin × $\frac{R}{R_1 + R_2}$ and substitute.
- For sensor questions, state the change of resistance, then the change of its fraction of the total resistance, then the change of the output p.d.

- For loading, combine the load and output resistor in parallel first, then recalculate the ratio.
- Give the voltage in V to 3 s.f.; resistances in kΩ may be used directly if they are consistent, as the ratio is dimensionless.

Section A traps

Trap: Using the wrong resistor in the numerator: the output across R_2 is Vin $\frac{R_2}{R_1 + R_2}$, not Vin $\frac{R_1}{R_1 + R_2}$.

Trap: Believing that the output voltage remains the same when a load is connected; the parallel load decreases it.

Trap: Believing that an LDR in bright light has an increased resistance; its resistance decreases, so the p.d. across it decreases.

Trap: Using $\frac{Vin}{R_1 + R_2}$ without multiplying by R, giving a current instead of a voltage.

Tricks

Trick: Identical resistors in series share the supply equally: N equal resistors give $\frac{Vin}{N}$ across each.

Trick: Sensor in the lower position: when its resistance falls, the output falls; swapping the positions reverses the effect.

15.4 Potentiometer and Wheatstone Bridge

Definitions & laws

- **A potentiometer** is a device used to measure the e.m.f. of a cell, or to compare e.m.f.s, by balancing the p.d. across a length of uniform wire against the e.m.f. of the cell, so that no current is drawn from the cell at balance.
- **At the balance point** the galvanometer reads zero, so there is no current in the galvanometer branch and no p.d. across it.
- **The potentiometer** is more accurate than a voltmeter because at balance no current is drawn from the cell, so the measured value equals the e.m.f. and not the terminal p.d.
- **A Wheatstone bridge** is balanced when the galvanometer reads zero, which occurs when the ratio of the resistances in one arm equals the ratio in the other, $\frac{P}{Q} = \frac{R}{S}$.
- **At balance** the two points connected to the galvanometer are at the same potential.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Potentiometer balance length	$E \propto l;\ \frac{E_1}{E_2} = \frac{l_1}{l_2}$	l = balance length (cm or m); for a uniform wire with a constant driver current
E.m.f. from balance length	$E = \frac{l}{L} V_0$	L = total wire length (same unit as l); V_0 = p.d. across the whole wire (V)
P.d. across the wire	$V_0 = IR_{wire}$	I = driver current (A); R_{wire} = resistance of the wire (Ω)
Driver circuit current	$I = \frac{E_{driver}}{R + R_{wire} + r}$	E_{driver} = e.m.f. of the driver cell (V); R = series resistance (Ω); r = internal resistance (Ω)
Wheatstone bridge balance	$\frac{P}{Q} = \frac{R}{S}$	P, Q, R, S = resistances of the four arms (Ω); at balance
Slide-wire bridge	$\frac{R}{X} = \frac{l_1}{l_2}$	X = unknown resistance (Ω); l_1, l_2 = lengths of the wire on either side of the balance point; resistance is proportional to length for a uniform wire

How to answer (MPM style)

- State that E is proportional to the balance length because the wire is uniform and carries a constant current.
- Write $\frac{E_1}{E_2} = \frac{l_1}{l_2}$, substitute the lengths and then calculate the unknown e.m.f. with unit.
- Explain zero deflection by stating that there is no current through the galvanometer, hence no current drawn from the cell.
- For the Wheatstone bridge, write the balance condition first, identify the correct arms, then solve for the unknown resistance.

Section A traps

Trap: Inverting the ratio: the cell with the longer balance length has the larger e.m.f., so $\frac{E_1}{E_2} = \frac{l_1}{l_2}$, not $\frac{l_2}{l_1}$.

Trap: Writing P + Q = R + S, P × Q = R × S or P − Q = R − S for the bridge condition; the correct condition is a ratio.

Trap: Using the wrong arm: on a slide wire the unknown resistor is in the ratio with the length on its own side, and its position determines whether R is smaller or larger than the standard.

Trap: Omitting the series resistor when calculating the driver current, so that the p.d. across the slide wire is wrongly taken as the driver e.m.f.

Tricks

Trick: Sanity check: the unknown cell with the shorter balance length must have the smaller e.m.f.; eliminate options that contradict this.

Trick: In a balanced bridge cross-multiply: P × S = Q × R, which is the same as $\frac{P}{Q} = \frac{R}{S}$; the products of the opposite arms are equal.